CROSSSAFE · TRACEABILITY MATRIX

CrossSafe traceability matrix

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/crosssafe		
Source	docs/03b-traceability.md @ d9f28be		

REVISION HISTORY

Version	Date	Author	Change
0.5	2026-10-31	Amish Chadha	Generated by .kit/trace.py
0.4	2026-10-31	Amish Chadha	Generated by .kit/trace.py
0.3	2026-10-31	Amish Chadha	Generated by .kit/trace.py
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CrossSafe traceability matrix

Generated by `python .kit/trace.py` from the user requirements, product requirements, calculations, build plan and BOM. Do not edit by hand. 14 user requirements, 19 product requirements, 0 gaps.

Table 1. User requirement to product requirement to evidence to verification.

User requirement	Product requirement	Title	Evidence	Verification	Status
UR-001, UR-006	PR-001	Light geometry and pattern	CRS-CAL-001, CRS-BLD-001	Design review against the pilot jurisdiction's rules	Met by design
UR-001, UR-014	PR-002	Visibility	CRS-CAL-001, CRS-STM-001, CRS-OPT-001, BOM	Photometric calculation from LED data; later field check	Not verifiable at TRL 3
UR-002	PR-003	Activation and sync	CRS-CAL-001	Timing budget; later bench test	Met on paper
UR-004	PR-004	Flash duration	CRS-CAL-001, CRS-BLD-001	Firmware review	Met by design
UR-002	PR-005	Detection quality	CRS-CAL-001, CRS-BLD-001, CRS-OPT-001	Site trial with manual counts	Not verifiable at TRL 3
UR-005	PR-006	Autonomy without sun	CRS-CAL-001, CRS-BLD-001, CRS-STM-001	Energy calculation	At risk
UR-005	PR-007	Energy balance	CRS-CAL-001, CRS-BLD-001	Solar yield calculation	Met on paper
UR-006	PR-008	Environment	CRS-CAL-001, CRS-BLD-001, CRS-STM-001, BOM	Datasheets and heat balance; later spray and thermal tests	Met on paper
UR-006	PR-009	Structure	CRS-CAL-001, CRS-BLD-001, CRS-STM-001	Wind load calculation	Met on paper
UR-008	PR-010	Mounting	CRS-CAL-001, CRS-BLD-001, CRS-STM-001	Design review; later timed trial	Not verifiable at TRL 3
UR-003	PR-011	Accessibility	CRS-CAL-001, CRS-BLD-001	Design review with disability groups	Met by design
UR-007	PR-012	Fail-safe behavior	CRS-CAL-001, CRS-BLD-001	Firmware review; fault injection later	Met by design
UR-011	PR-013	Privacy	CRS-CAL-001	Design review	Met by design
UR-012	PR-014	Theft and vandal resistance	CRS-CAL-001, CRS-BLD-001	Design review	Met by design
UR-009	PR-015	Affordable	CRS-CAL-001, CRS-OPT-001	Priced BOM (`bom/bom.csv`)	Over target
UR-013	PR-016	Flash area	CRS-CAL-001, CRS-OPT-001	Analysis (CAL-001 section I); a road-authority and clinical-safety review is to assess	At risk

User requirement	Product requirement	Title	Evidence	Verification	Status
UR-006	PR-017	Light bar temperature	CRS-CAL-001, CRS-STM-001	Analysis (CAL-001 C5); datasheet of the heads bought	Met on paper
UR-014	PR-018	Warning in time to stop	CRS-CAL-001, CRS-STM-001	Analysis (CAL-001 section I); AASHTO values to confirm	Met on paper
UR-010	PR-019	Buildable and repairable	CRS-CAL-001	Build plan BLD-001 (paper); model of 117 parts, none bare (CAL-001)	Met by design

Gaps

None.